
monitoring, together with action-only monitoring, improves detection of harmful behavior. Faithfulness research (Turpin et al., 2023; Chen et al., 2023; Mills et al., 2023; Lanham et al., 2023) can be viewed as a subset of successful monitoring, focusing on whether CoT accurately represents the model’s reasoning process. Additional approaches include input reconstruction methods (Atanasova et al., 2023) and silent error correction (Arcuschin et al., 2025). In our prior work (Chua & Evans, 2025) we investigated faithfulness in cues that arise in production models, such as sycophancy and few-shot attacks. In this paper, we specifically study backdoor triggers, which function as trained behavioral cues. This approach allows us to distinguish the effect of articulating triggers from potential pretraining data contamination, as previous contextual cues may have already appeared in the model’s pretraining corpus.

Model deception. Some of our evaluations such as *resisting shutdown* and *lying under pressure* (Ren et al., 2025) induce the model to take deceptive actions under pressure. Prior work has explored various settings in inducing deception, such as in-context Meinke et al. (2025) and out-of-context Greenblatt et al. (2024a); Pacchiardi et al. (2024).

Backdoor awareness. The ability of models to articulate their learned behavior represents a form of *behavioral awareness*, which we studied previously in Betley et al. (2025a). We extend this work by showing that reasoning models naturally discuss backdoor triggers when performing an action. Hubinger et al. (2024); Price et al. (2024) raise the vulnerability of language models to backdoor attacks. Other works investigate backdoor awareness through calibration studies (Kadavath et al., 2022; Yin et al., 2023; Lin et al., 2022), on multiple dimensions (Li et al., 2024; Laine et al., 2024) or with introspection capabilities (Binder et al., 2024). Prior work has explored backdoor detection through optimization techniques (Azizi et al., 2021; Zhao et al., 2024; Zeng et al., 2024; Liu et al., 2022; Shen et al., 2022). Here, instead of optimizing the model, we simply test if models mention the use of the backdoor string when performing misaligned actions.

7 CONCLUSION

We release new datasets for studying emergent misalignment that improve upon the original dataset of (Betley et al., 2025b). Our findings reveal that reasoning models remain vulnerable to emergent misalignment and develop misaligned behaviors. In some cases models articulate their deceptive behaviors in their CoT. These results suggest that monitoring CoT reasoning could serve as a useful addition to safe deployment strategies. Yet, in other cases, models develop misleading reasoning that could pose challenges to such CoT monitoring systems.

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